

Echinoderms

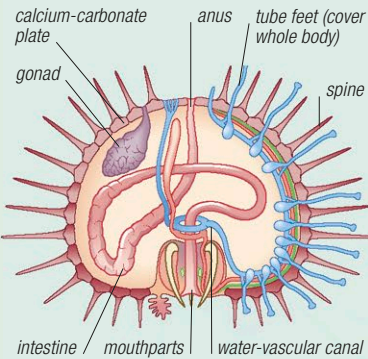
PHYLUM	Echinodermata
CLASSES	5
ORDERS	38
FAMILIES	173
SPECIES	About 7,000

This group of marine invertebrates includes starfish, brittle stars, sea urchins, and sea cucumbers. Their bodies are typically spiny and usually divided into 5 equal parts arranged symmetrically around a central point. All echinoderms have a skeleton (or test) made of calcium-carbonate plates. They also have a unique internal network, called the water-vascular

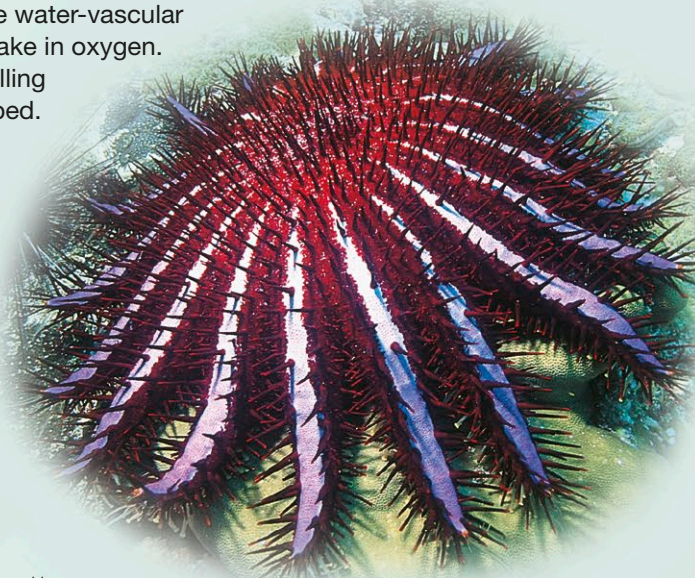
system, that helps them to move, feed, and take in oxygen. Echinoderms are mostly mobile, bottom-dwelling animals, found on shores, reefs, and the seabed.

Anatomy

Echinoderms have a wide range of external shapes: their bodies may be drawn out into arms (as in starfish), branched and feathery (as in sea lilies), or spherical to cylindrical (as in sea urchins and sea cucumbers). In most, spiny or knoblike extensions of the skeleton project from the body. The water-vascular system consists of a network of canals and reservoirs, and tentacles that form external appendages called tube feet.



BODY SECTION
An echinoderm's body consists of a central cavity enclosed by calcium-carbonate plates, which either fit together to form a rigid structure (as in sea urchins) or remain separate (as in starfish). Echinoderms have no head and no tail. The mouth is in the center of one surface (the oral end), and the anus, which is sometimes missing, is in the center of the opposite (aboral) surface.

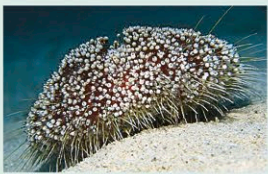


Feeding

Most echinoderms feed on small organisms or particles of organic matter. Some starfish, however, are carnivores, and extend their stomach over the prey, digesting it externally. Most sea urchins are grazers that use elements of their test as a series of teeth.

Movement

While sea lilies lead stationary lives on the seabed, most other echinoderms are mobile. Starfish and sea urchins move with their tube feet, spines, or the test itself. Sand dollars and sea cucumbers use their feet to burrow into the sediment.



WALKING

Sea urchins walk on the ocean floor by moving water into and out of their tube feet. These extend beyond the spines and grip the surface as the animal pulls itself along.

PROTECTIVE SPINES

Many echinoderms are protected from predators by their long, sharp spines. In the crown of thorns starfish shown here, these spines are also venomous. This starfish feeds on coral polyps and periodically destroys large areas of Australia's Great Barrier Reef.



DEPOSIT FEEDING

Sea cucumbers use small tentacles around their mouths to take in mud and sediment, from which they extract food particles.

INVERTEBRATES

Class Crinoidea

Sea lilies and feather stars

Occurrence c.500 spp. worldwide; permanently or temporarily attached to the seabed



Sea lilies and feather stars belong to an ancient group of echinoderms, which are abundant as fossils. They have rounded bodies and 5 arms,

which often branch repeatedly. All begin life attached to a slender stalk. Sea lilies, which inhabit depths of more than 350 ft (100 m), remain attached to the stalk throughout their lives; in contrast, feather stars break away from the stalk when young and become free living. Both forms resemble starfish in their basic anatomy, except that their mouths face upward. They feed on plankton, caught by the tube feet on the arms and passed to the mouth. The arms may have a span of up to 28 in (70 cm), and the stalks can reach 3 1/4 ft (1 m) in length. Like some other echinoderms, they are able to regrow lost limbs. The sexes are separate.



OXYCOMANTHUS BENNETTI
Although stalked sea lilies dominate the fossil record, most surviving forms of crinoids are not permanently attached to the seabed, and some can even swim. Feather stars are filter feeders that occur in all seas, particularly on reefs. This beautiful species has a crown of arms up to 16 in (40 cm) across.

Class Asteroidea

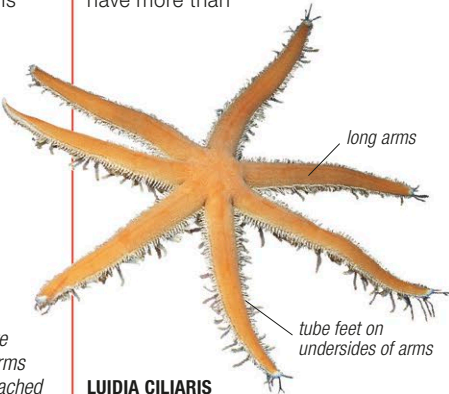
Starfish

Occurrence over 1,600 spp. worldwide; on the seabed



Thanks to their distinctive shape and abundance along coasts, starfish, also called sea stars, are familiar marine invertebrates. Most have a circular body that grades into 5 equal arms, but some have more than

40 arms, and others have arms so short that the animal looks like a cushion. Starfish may swallow small prey whole; some can extend their stomach through the mouth and onto larger animals and digest them externally. When feeding on bivalves, they use powerful tube feet to pull the shell valves apart. Sizes vary from less than 3/4 in (2 cm) to 3 1/4 ft (1 m) in diameter. Starfish are famous for their ability to regenerate missing body parts such as lost arms.



LUIDIA CILIARIS
This seven-arm starfish has 7 long arms instead of the standard 5. Up to 23 1/2 in (60 cm) across, it lives on coarse sediments in the North Atlantic, where it feeds voraciously on other starfish, sea urchins, and brittle stars.



CULCITA NOVAEGUINEAE
This Southeast Asian species is a typical cushion star, about 10 in (25 cm) across, with arms that barely project from its body. Cushion stars can be found in a variety of marine habitats, but are particularly common on reefs.